

Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend engineer and team lead with 6+ years building event-driven microservices and distributed systems — Apache Kafka and Google Pub/Sub messaging, Kubernetes and Docker, across GCP and AWS. Led a team of 4 engineers delivering production systems end to end, and led the zero-downtime migration of 20+ production applications reporting directly to the CTO — owning the full software lifecycle from architectural direction through production support and decommissioning. Polyglot backend developer (JavaScript/TypeScript, Python, Go) with a record of mastering new stacks fast: joined MTailor as a contractor on an unfamiliar platform and within months was leading the company-wide migration and writing the docs other engineers onboarded from. Based in Kampala (UTC+3); built USSD services for underserved users in Uganda — now applying event-driven architecture and servant leadership to financial inclusion at scale.

CORE COMPETENCIES

Team Leadership & Mentoring

Event-Driven Architecture

Microservice Development

Kafka & Pub/Sub Messaging

Kubernetes & Docker

GCP & AWS Cloud

Zero-Downtime Migrations

Full Lifecycle Ownership

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote

- **Led the end-to-end migration of 20+ production applications** from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to the CTO — setting technical direction and owning the full software lifecycle from design through production support and decommissioning
- Built a **real-time, event-driven two-way sync** between MongoDB and Firestore using Node.js and Google Pub/Sub with idempotent message processing, keeping two live datastores consistent under parallel production traffic
- Cut infrastructure spend by **\$5,000/month** by migrating all services off AWS; migrated file storage from Amazon S3 to GCS via a Python pipeline
- Implemented Express Shipping in partnership with product priorities, generating an additional **\$40 revenue per order**; built a 3D visualisation feature with video overlay (ffmpeg) that increased buyer conversion
- Migrated a Webflow website from a Parse backend to Firebase

MTailor

Mar 2022 – Jul 2022

Contractor (Backend Migration) — US Remote

- Hired onto an unfamiliar Firebase/GCP stack and delivered migration scripts reading from MongoDB and writing to Firestore with complex processing logic — converted to full-time and given leadership of the company-wide migration
- **Mentored and enabled fellow engineers:** wrote the documentation for the new Firebase SDKs (Node.js and Python) that the team onboarded from; trained the Ops team on the new dashboard and backup procedures

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer — US Remote

- Built backends for DeFi (decentralized finance) applications using Web3 and Node.js, reporting directly to the CTO
- Delivered production React UIs from Figma designs under tight agency timelines

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Led optimisation of a PWA delivering real-time stock market prices from a Firebase backend
- Built batched jobs querying over **2 million Firestore records** to keep customer prices current from Morningstar APIs, optimizing query shape over hardware scale
- Extended backend APIs with Firebase Cloud Functions and Express.js

CodeBits

Jan 2020 – Jul 2021

Software Engineer (led team of 4) — Uganda Remote

- **Led a team of 4 developers** building legal tech systems for NGOs in Uganda — coaching through delivery, owning architectural guidance and technical direction
- **Architected an event-driven microservices backend** for FIDA Uganda's case management platform using Apache Kafka, Docker, and Kubernetes

- Built a **USSD service for legal aid providers** serving low-connectivity users, communicating with a Node.js backend via gRPC on a provider-agnostic, versioned contract
- Delivered a case management system for FIDA Uganda, a paralegal database for LASPNET, and a cross-platform React Native mobile app (iOS & Android) with Firebase push notifications

PROJECTS

Zero-Downtime Platform Migration (MTailor)

Migrated 20+ production apps from Parse/MongoDB to Firebase/GCP with zero downtime and recurring monthly infrastructure savings, using a phased cutover with rollback preserved for the entire migration window.

Node.js · GCP · Firebase · MongoDB · Pub/Sub

Real-Time Pub/Sub Sync Pipeline

Live two-way MongoDB ↔ Firestore sync with idempotent, event-driven message processing — supported parallel production traffic on two datastores with no data loss.

Node.js · Google Pub/Sub · Firestore · MongoDB

Healthcare WhatsApp Chatbot

Prescription OCR and patient assistance over WhatsApp Business API — integrating OpenAI and Google Vision to reach users on the messaging infrastructure they already use.

Node.js · WhatsApp Business API · OpenAI · Google Vision

EDUCATION

BSc Computer Engineering — [Makerere University](#), Kampala

2018 – 2022

LANGUAGES

English (fluent) Kiswahili

SKILLS

Languages: JavaScript/TypeScript (Node.js), Python, Go, SQL, Rust (basic)

Messaging & Streaming: Apache Kafka, Google Pub/Sub, NATS Streaming, gRPC

Cloud & Infrastructure: GCP, Firebase, AWS (S3/EC2/EBS), Kubernetes, Docker, Linux, Git **Data:** MongoDB, Firestore, SQL databases

Frontend & Mobile: React, React Native, Tailwind